

Course Title: INTRODUCTION TO MECHANICAL ENGINEERING

Course Code: 22ESC144/244

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Module 1

Introduction to Mechanical Engineering (Overview only)

Role of Mechanical Engineers in Industries and Society- Emerging Trends and Technologies in different sectors such as Energy, Manufacturing, Automotive, Aerospace and Marine sectors and contribute to the GDP

1.1 Role of Mechanical Engineers in Industries and Society

Mechanical engineering plays a critical role in manufactured technologies, from cars to airplanes to refrigerators. It enables you to do many daily activities with ease, as it brings helpful technologies to our modern society.

Mechanical engineers work in various industries, building and designing nuclear plants, or automobiles, or railway equipment, or spacecraft. These mechanical engineers are sometimes called nuclear engineers, automotive engineers, railway engineers, and aerospace engineers.

Mechanical engineers also work as consulting practices, government and universities. They may work in classrooms, factories offices, laboratories or testing facilities as teachers, managers, designers or researchers. Some mechanical engineers work in sales and product quality control or equipment maintenance. Many hold managerial positions in their companies. Whether working on an oil rig or in corporate headquarters, mechanical engineers are solving the technical problems of today and tomorrow.

1.2. Emerging Trends and Technologies in Energy sector

Renewable energy infrastructure development, power generation, storage, and efficiency drive innovations in the sector with numerous emerging companies developing low-cost renewable energy technology.

1) Renewable

Renewable is one of the emerging trends in energy sector. This Renewable energy trend helps to safeguard the environment by emitting little to no harmful pollutants. The fundamental premise of renewable energy is to obtain it from a consistent source in the environment, such as the sun, wind, or geothermal sources. The source is then converted into useful power or fuel.

A variety of technologies that address various facets of generating power and heat from renewable sources is one of the latest technology trends in renewable energy. This involves lowering the cost of building renewable infrastructure and enhancing the efficiency of power generation.

2) Internet of Energy (IoE)

In the traditional method, during the time of construction, the electric power system uses the central architecture and this poses a number of obstacles to the industry. And here IoE comes into the frame. It helps to recognize some of the challenges and accordingly propose the best efficiency and optimal design to build an energy system.

Technology trends in the energy industry refer to automate the power infrastructure in order to manufacture and produce energy

Benefits:

- Reduce insufficiency
- Make the transmission of energy more productive
- Reduce the expenses
- Decrease in the wastage of energy

3) Energy Storage

Energy storage is one of the effective energy industry emerging trends. Energy storage permits steady pricing by proactively maintaining demand from consumers

In response to shifting energy demands and technological advancements, the energy storage business has evolved, adapted, and innovated during the last century. Energy storage systems offer a diverse set of technological solutions for improving our power supply to build more resilient energy infrastructure and save money for utilities and customers.

The many methods of energy storage can be classified into five categories based on their technology:

- Batteries
- Thermal
- Mechanical
- Pumped hydro
- Hydrogen

It has the potential to save consumers money while also improving dependability and resilience, integrating power sources, and reducing environmental impacts

4) Block chain

Blockchains or distributed ledgers are emerging technology trends in the energy industry that has drawn significant interest from energy supply firms, start-ups, financial institutions, governments, technology developers, and the academic community.

Block chain technology proposes to combine all energy stakeholders under a single decentralized system. Electricity producers, metering operators, distribution network operators, and traders potentially avail from utilizing smart contracts. These agreements ensure that all energy-related transactions are routed through a safe and immutable network, reducing the risk of losses. Block chain also has the ability to achieve some degree of equity between energy generators and consumers

1.3 Emerging Trends and Technologies in manufacturing sector

AI, robots, 3D printing, and the like are all the latest manufacturing trends in manufacturing technology – and have been for some time. 2021 is the year manufacturing technology gets even smarter. Robots on the factory floor are growing at warp speed thanks to their declining cost and increasing capabilities

1.4 Emerging Trends and Technologies in Automotive sector

The automotive industry has been experiencing a great deal of change within the past several years. More software is being added to vehicles, along with a greater number of electric and autonomous vehicles are in production and on the road.

Here are the most common automotive trends of 2021.

- A Greater Amount of AI/Machine Learning and Connectivity
- An Increase In Electric and Autonomous Vehicles
- Software Security Has Become a Top Concern
- Electrification
- Artificial Intelligence (AI)
- Human-Machine Interface

1.4 Emerging Trends and Technologies in Aerospace sector

Aerospace collectively refers to the atmosphere and outer space; it's a diverse industry with a multitude of commercial, industrial, and military applications. Aerospace engineering consists of aeronautics and astronautics and the research, design, production, operation, or maintenance of aircraft and spacecraft involves the work of numerous organizations.

10 Aerospace Engineering Technology Trends

1. Zero-Fuel Aircraft
2. Structural Health Monitoring (SHM)
3. Advanced Materials

4. Smart Automation and Block chain
5. Additive Manufacturing (3D Printing)
6. Supersonic Flights
7. More Resilient and Dynamic A&D Supply Chains
8. Utilizing Internet of Things (IoT) to Anticipate Maintenance Issues
9. Artificial intelligence (AI)
10. Autonomous Flight Systems

What is emerging technologies in the maritime industry?

Advances in shipbuilding, propulsion, smart shipping, advanced materials, big data and analytics, robotics, sensors and communications in conjunction with an increasingly skilled workforce are all having monumental shifts in how the maritime industry are approaching new challenges and opportunities.

The Indian Manufacturing sector currently contributes 16-17% to GDP and gives employment to around 12% (2014) of the country's workforce. Various studies have estimated that every job created in manufacturing has a multiplier effect in creating 2–3 jobs in the services sector.

Petroleum based liquid fuels.

Petrol : Known as gasoline. It is a mixture of hydrocarbon. It is the lightest and most volatile liquid fuel mainly used as a motor fuel.

Petrol is obtained from distillation of crude petroleum. It is mainly used in S.I. Engines.

Diesel oil: The liquid oil obtained after the distillation of gasoline and kerosene is known as Diesel oil. It evaporates slower than gasoline & is heavier. Diesel is used as a fuel in Diesel engines and to run variety of engines.

Kerosene: It is also known as paraffin oil. It is a combustible hydrocarbon liquid fuel. It is obtained from crude petroleum in the fractional distillation after obtaining petrol. It is heavier than petrol and less volatile than petrol. It is mostly used as domestic fuel for heating and lighting purpose.

Petroleum based Graseous fuels.

Natural Gas: It is extracted from the underneath earth surface by deep drilling. Methane is the main constituent of natural gas and the other elements are Ethane, propane, Butane, CO₂ and traces of other gases.

Natural Gas has high calorific value. It mixes with air readily and does not produce smoke. It is lighter than air and disperses into air easily in case of leakage. It is used in domestic, commercial, industry and transportation sector.

Liquified Petroleum Gas (LPG)

It is a mixture of ethane, propane and butane. The petroleum gas which is liquified under pressure is called LPG. It is produced as a by product of natural gas purification and oil refining.

on compression, it easily changes to a liquid state, which is a unique and special property of it. due to this it occupies a small space. $\therefore$ it is stored in cylinders and transported.

It is extensively used as a domestic fuel and in transportation.

Compressed Natural Gas (CNG)

CNG is basically natural gas which is obtained from petroleum products. It consists of methane. It can be used in place of petrol, diesel and LPG. It is non-toxic, colourless & lighter than air. It is safer than other fuels.

Fossil fuels.

4

Energy is being used in many ways throughout the day. furnaces and stoves use thermal energy to heat buildings & cook food. Air Conditioners use electrical energy to move thermal energy outdoors. Cars and other vehicles use mechanical energy to carry people and materials from one part of the Country to another.

Energy can only be transformed or converted from one form to another.

* Fossil fuels are formed from the decaying of plants and animals. Ex: Petroleum or oil or natural gas or Coal.

Fossil fuels contain high percentages of carbon and include Petroleum, coal & natural gas

Petroleum. (oil)

Millions of gallons of petroleum or crude oil are pumped everyday from wells deep in Earth's crust.

Petroleum is a highly flammable liquid formed by decayed ancient organisms such as microscopic plankton and algae. It contains only carbon & hydrogen.

Applications: All the major means of transportation depend on fossil fuels. Petroleum products such as Gasoline, Diesel, Kerosene are used to power the engines of millions of vehicles. Even Aeroplanes and ships use petroleum based transportation fuels.

COAL

A significant amount of fossil fuel i.e. coal is used for generating electricity. Coal is the source of fuel in maximum number of thermal power plants, water is converted into steam by the heat generated by burning

Natural gas.

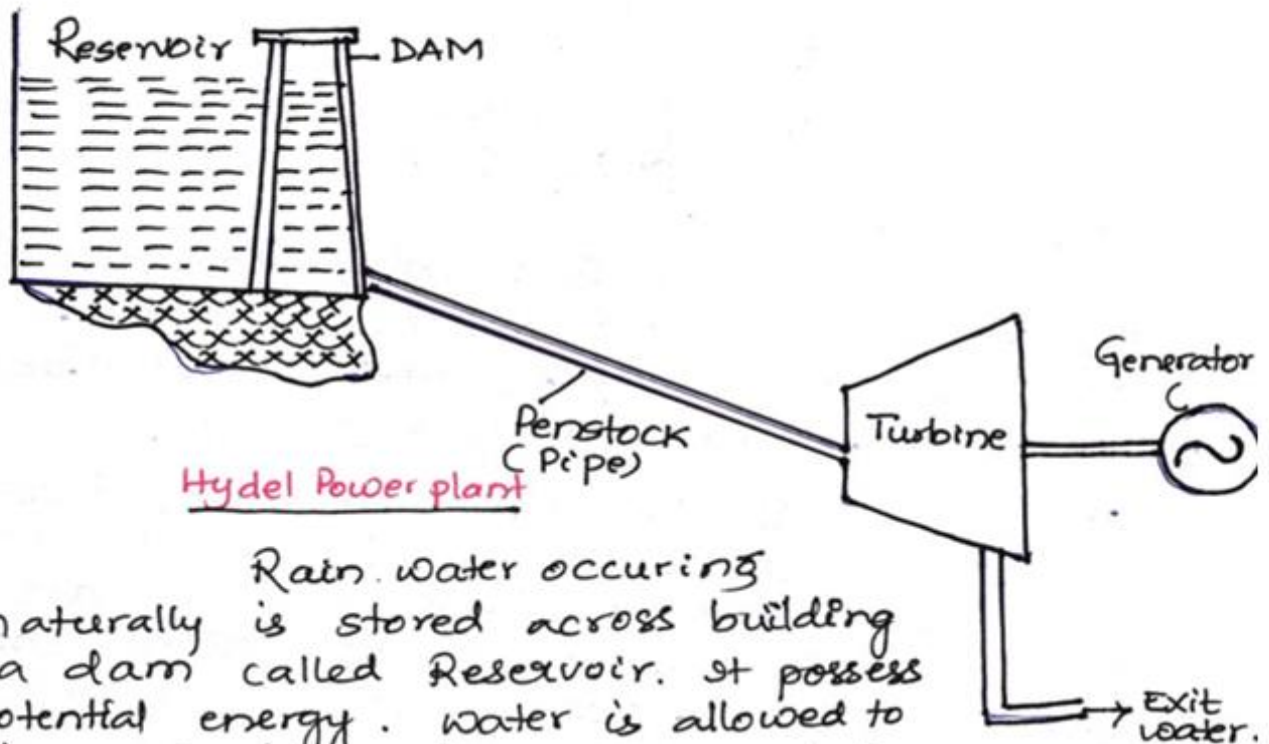
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The final variety of fossil fuel is natural gas. It is gaseous in nature and made up of primarily methane and is light in weight as well flammable.

Natural gas is mainly used to heat homes, power air conditioning systems, fuel stoves and other cooking appliances.

HYDEL ENERGY.

When electricity is generated from flowing water, it is called hydroelectric energy. It is considered as renewable energy resource.



Rain water occurring naturally is stored across building a dam called Reservoir. It possess potential energy. Water is allowed to

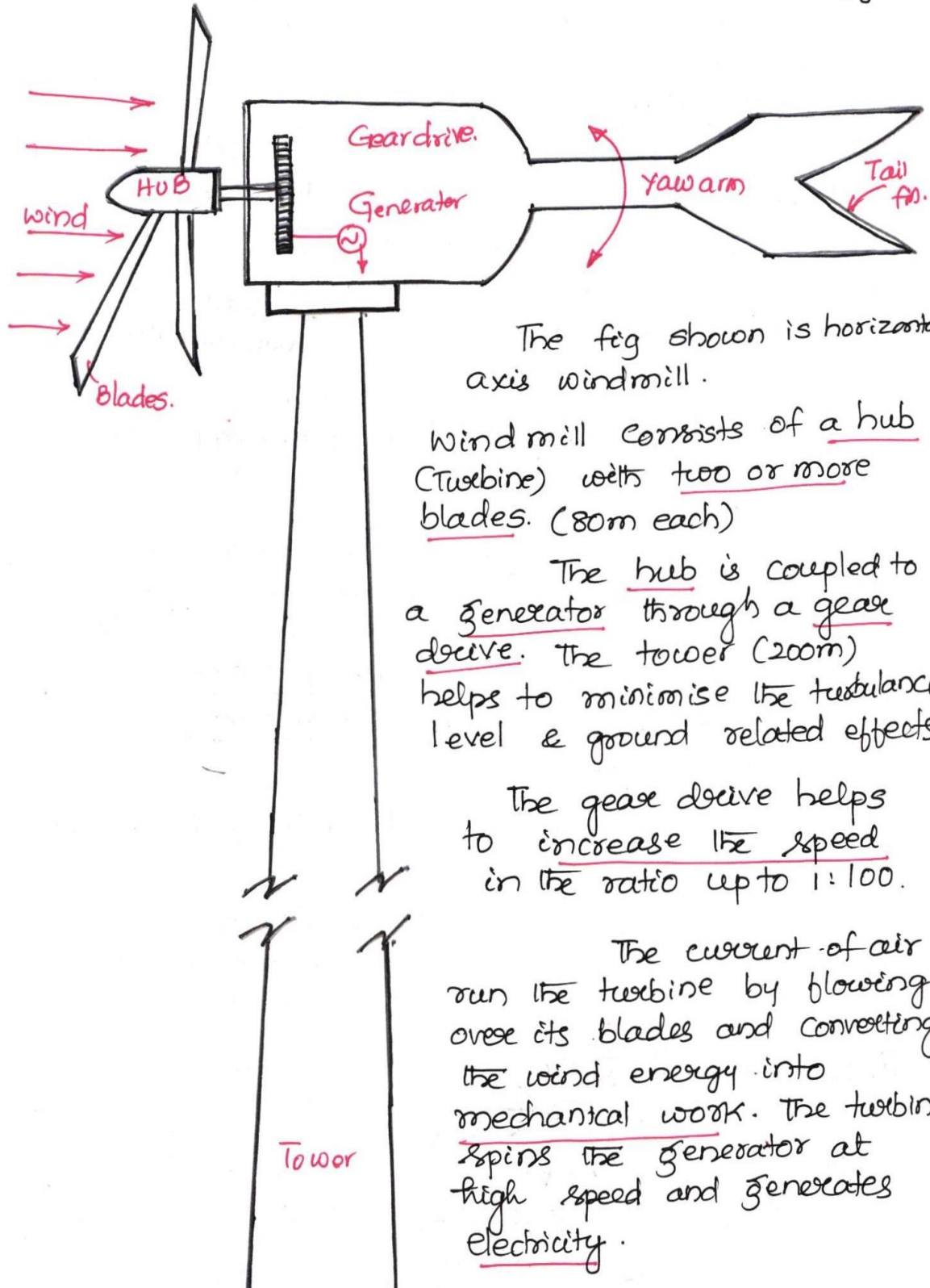
flow through a long pipe called penstock in which P.E of water is converted into kinetic energy and impinges over blades of rotating turbine due to which hydel energy is converted into rotational energy and then into mechanical work. A generator coupled to turbine converts mechanical work into electrical energy and the water exits the turbine to other purposes like agricultural fields and rivers.

ADVANTAGES: - Environmental friendly source, large scale power generation, energy at free of cost

DISADVANTAGES: - Expensive to build the dam, summer water may not sufficient to produce electricity

Wind energy.

It is the Kinetic energy of air in motion
wind is an indirect form of solar energy.



The fig shown is horizontal axis windmill.

Windmill consists of a hub (Turbine) with two or more blades. (80m each)

The hub is coupled to a generator through a gear drive. The tower (200m) helps to minimise the turbulence level & ground related effects.

The gear drive helps to increase the speed in the ratio up to 1:100.

The current of air runs the turbine by blowing over its blades and converting the wind energy into mechanical work. The turbine spins the generator at high speed and generates electricity.

ADVANTAGES:-

1. The wind is free and with modern technology it can be captured efficiently. 2. Once the wind turbine is built the energy it produces does not cause greenhouse gases or other pollutants.
3. Many people find wind farms an interesting feature of the landscape
4. Remote areas that are not connected to the electricity power grid can use wind turbines to produce their own supply.
5. Wind turbines are available in a range of sizes which means a vast range of people and businesses can use

Disadvantages:-

1. Wind turbines are noisy
2. The strength of the wind is not constant and it varies from zero to storm force.
3. Only selected places it can be harnessed.

NUCLEAR FUELS

Nuclear energy is the energy that holds the nucleus of an atom. The energy released during nuclear fission or fusion, especially when used to generate electricity.

Nuclear fusion: Fusion energy is a form of nuclear energy released by the fusion (combustion) of two light nuclei (i.e. nuclei of low mass) to produce heavier mass.

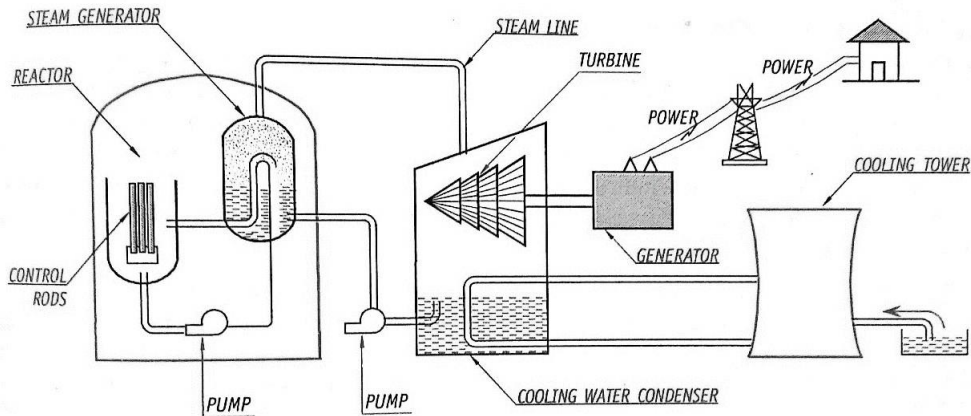
Nuclear fission: Nuclear fission is the process, where a heavy nucleus splits into two fragments of more or less of equal mass.

Nuclear Reactor

A nuclear reactor is a device which controls the nuclear fission chain reaction to harness nuclear energy for peaceful purposes. A nuclear reactor which is used to generate electricity, is called a nuclear power plant. Fuel in the form of pellets is enclosed in several tubular claddings of steel or aluminum. This is called fuel assembly. Enriched U-235 or Pu-239 is the fuel material. A coolant is circulated through the reactor to remove the heat generated. Ordinary water is most commonly used coolant. Rods made of boron or cadmium which are neutron absorbers are used as control rods. The neutrons available for fission are controlled by moving the control rods in and out of the nuclear core. The rods can be used to shut down the reactor.

Working principle of a nuclear power station

The schematic diagram of nuclear power station is shown in figure. A generating station in which nuclear energy is converted into electrical energy is known as nuclear power station. The main components of this station are nuclear reactor, heat exchanger or steam generator, steam or gas turbine, AC generator and exciter, and condenser.



Nuclear Energy Conversion

The reactor of a nuclear power plant is similar to the furnace in a steam power plant. The heat liberated in the reactor due to the nuclear fission of the fuel is taken up by the coolant circulating in the reactor. A hot coolant leaves the reactor at top and then flows through the tubes of heat exchanger and transfers its heat to the feed water on its way. The steam produced in the heat exchanger is passed through the turbine and after the work has done by the expansion of steam in the turbine, steam leaves the turbine and flows to the condenser. The mechanical or rotating energy developed by the turbine is transferred to the generator which in turn generates the electrical energy and supplies to the bus through a step- up transformer, a circuit breaker, and an isolator. Pumps are provided to maintain the flow of coolant, condensate, and feed water.

ADVANTAGES:-

Nuclear power generation does emit relatively low amounts of carbon dioxide (CO₂).

The emissions of greenhouse gases and therefore the contribution of nuclear power plants to global warming is therefore relatively little.

This technology is readily available, it does not have to be developed first.

It is possible to generate a high amount of electrical energy in one single plant.

DISADVANTAGES:-

The problem of radioactive waste is still an unsolved one.

High risks: It is technically impossible to build a plant with 100% security.

The energy source for nuclear energy is Uranium. Uranium is a scarce resource, its supply is estimated to last only for the next 30 to 60 years depending on the actual demand

SOLAR ENERGY.

Energy radiating from the sun in the form of heat and light is called Solar energy. It is conventional as well renewable energy source. Solar energy reaches the earth @ 1.353 kW/m^2 and is called Solar constant.

only about 3% of solar energy is being used worldwide due to reasons that

- It is not constantly available.
- It is not constantly available.
- It is difficult to convert
- It is scattering.

Solar panels, called photovoltaic cells are used to convert the sun's energy into electricity.

Solar energy can be converted directly into thermal energy or electrical energy.

Thermal energy can be utilized for a variety of applications such as heating water, drying, heating or cooling space, refrigeration and air conditioning.

Utilisation of solar energy is grouped under three primary processes.

1. Helio chemical process. (Photosynthesis)
2. Helio Electrical process. (Photovoltaic effect)
3. Helio thermal process. (Direct heating)

3. Heliothermal Process

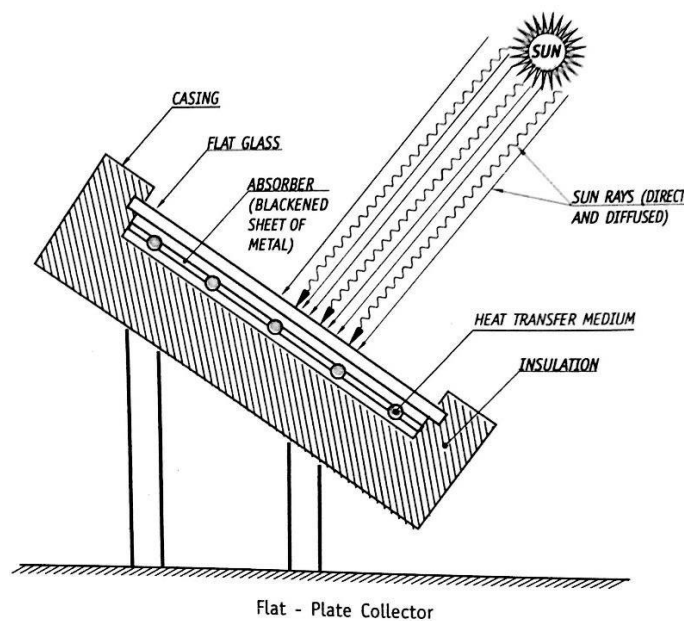
Radiant solar energy is directly converted into thermal energy (heat energy) by using a collector. This process is called as Helio thermal process. The surface on which the solar rays fall is called a collector. The collector may be either flat plate collector or focussing collector.

The surfaces on which the solar rays fall are called collectors.

There are two types of collectors:

(a) Flat plate collectors (b) Focusing collectors.

a) Liquid Flat Plate Collectors:



It has the following components:

(a) Absorbing plate

- Made of Copper, Aluminium or steel.
- It is coated with material to enhance the absorption of solar radiation.
- From the absorbing plates heat is transferred to tubes which carry either water or air.

(b) Transparent covers

- Sheets of solar radiation transmitting materials placed above the absorbing plate.
- They allow solar energy to reach the absorbing plate while reducing convection, conduction and re-radiation heat losses. Made of a toughened glass, usually 4mm thick. Which helps in reflecting the incident
- Solar energy back to the absorber plate.
- Glass cover permits the entry of solar radiation as it is transparent for incoming short wave lengths

(c) Insulation

It minimizes and protects the absorbing plate from heat losses.

Working

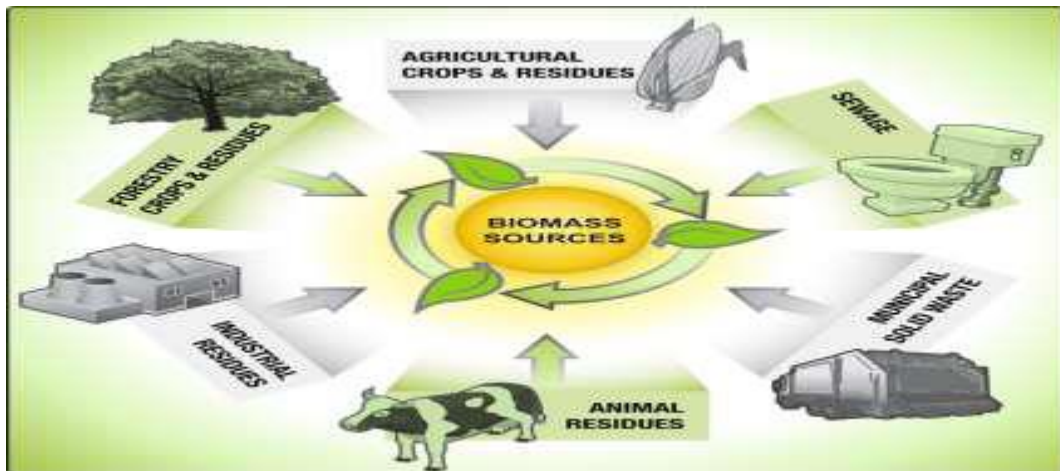
- Sun's rays falling on the transparent covers are transmitted to the absorbing plate.
- The absorbing plate usually of Cu, Al or galvanized iron is painted dead black for maximum absorption.
- The collector (plate) will absorb the sun energy and transfer it to the fluid in the pipe beneath the collector plate.
- Use of flat mirrors on the sides improves the output.
- Water from the overhead tank is made to flow through the water tubes. Solar rays passes through the transparent cover and falls on the absorber plate. Heat energy from the absorber plate is transferred to the cold water flowing through the tubes. Warm water rises above the cold water because of low density and flows into the heater tank.

Biofuels

Introduction: - Biomass is biological material derived from living, or recently living organisms. It most often refers to plants or plant-derived materials. As an energy source, biomass can either be used directly via combustion to produce heat, or Indirectly after converting it to various forms of biofuel.

This biomass may be transformed by physical, chemical and biological processes to biofuels. In chemical forms biomass is stored solar energy and can be converted into solid, liquid and gaseous energy carries. Biomass is biological/organic material derived from living, or recently living organisms. (The term is equally applicable to both animal and vegetable derived material, but in the context of energy, it refers to plant based material)

Source of biomass: - sources of biomass/biofuels are:-



Examples of various biofuels used in engineering applications: - The various bio-fuels are bio-methanol, bio-ethanol, bio-diesel, bio-gas and producer gas.

First-generation biofuels are made from sugar, starch, vegetable oil, or animal fats using conventional technology. The basic feedstock for the production of first-generation biofuels come from agriculture and food processing.

Introduction to Biogas Energy

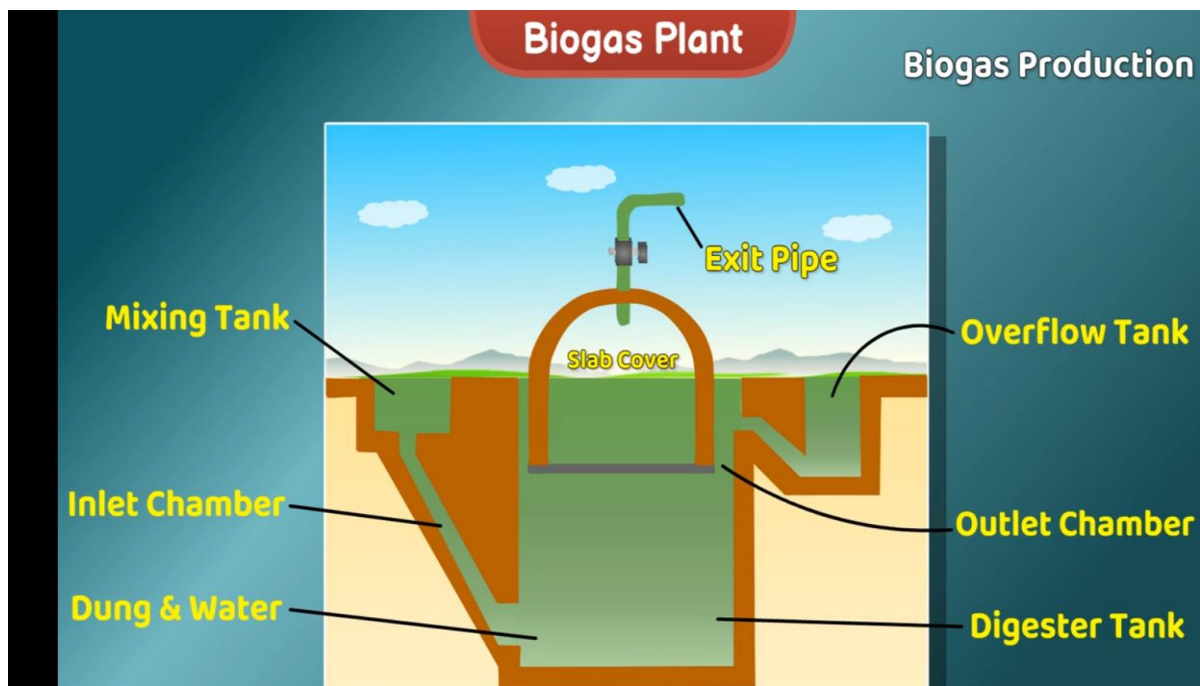
Biogas is a mixture of gases which the anaerobic decomposition of organic matter produces, for example, agricultural waste, plant residue, municipal waste, food waste and more. It comprises methane, carbon dioxide in conjunction with the small amount of hydrogen sulphide, and moisture.

Biogas Plant

The biogas plant comprises a dome-like structure. In this, the organic material like discarded food residue, fats, sludge, cow dung etc. mix with water and then fed into the digester through the inlet.

The digester is basically a sealed chamber where anaerobic decomposition of organic matter occurs. After a few days, the organic matter completely decomposes to produce gases like methane, carbon dioxide, hydrogen and hydrogen sulphide.

After that, we draw these gases through pipes from the storage tank above the digester and distribute through decentralization channels to neighbouring centres for use.



The most common first-generation biofuels are:

- Biodiesel: Extraction with or without esterification of vegetable oils from seeds of plants like soybean, oil palm, oilseed rape and sunflower or residues including animal fats derived from rendering applied as fuel in diesel engines.
- Bioethanol: Fermentation of simple sugars from sugar crops like sugarcane or from starch crops like maize and wheat applied as fuel in petrol engines.
- Biogas: Anaerobic fermentation of organic waste, animal manures, crop residues and energy crops applied as fuel in engines suitable for compressed natural gas.

Advantages of biomass energy:-

- Renewable and inexhaustible (theoretically) source of energy.
 - Biomass is very abundant.
 - It is easy to convert to a high energy portable fuel such as alcohol or gas which are efficient, viable and relatively clean-burning.
- It is cheap in contrast to the other energy sources.
 - Biomass production can often mean the restoration of waste land. Commercial use of biomass may reduce the problem of waste disposal.
- It may also use areas of unused agricultural land and provide jobs in rural communities.
 - When direct combustion of plant mass is not used to generate energy (i.e. fermentation, pyrolysis, etc. are used instead), there is minimal environmental impact.

Disadvantages of Biogas

In addition to having many advantages, there are also some disadvantages of biogas. They are as follows:

Inefficient on a large scale: As it is difficult to boost the efficiency of biogas, it is not economically feasible to use biogas on a large scale.

Contains impurities: It comprises a lot of impurities which are difficult to control even after putting it through many rounds of purification. When we compress biogas to use as fuel, it proves to be extremely corrosive to the container.

Unstable and hazardous: When methane comes in contact with oxygen it reacts violently to create carbon dioxide. As a result, the highly inflammable nature of methane makes it susceptible to explosions.

Global Warming:

Global warming is a gradual increase in the overall temperature of the earth's atmosphere generally attributed to the greenhouse effect caused by increased levels of carbon dioxide, CFCs, and other pollutants.

Causes for Global Warming:

One of the biggest issues facing us right now is global warming. Its effects on animals and on agriculture are indeed frightening, and the effects on the human population are even scarier. The causes are mainly from us, the human race, and the effects on us will be severe.

Carbon dioxide emissions from fossil fuel burning power plants Our ever increasing addiction to electricity from coal burning power plants releases enormous amounts of carbon dioxide into the atmosphere.

Carbon dioxide emissions from burning gasoline for transportation With our population growing at an alarming rate, the demand for more cars and consumer goods means that we are increasing the use of fossil fuels for transportation and manufacturing.

Methane emissions from animals, agriculture such as rice paddies, and from Arctic sea beds

Methane is another extremely potent greenhouse gas, ranking right behind CO₂. When organic matter is broken down by bacteria under oxygen-starved conditions (anaerobic decomposition) as in rice paddies, methane is produced.

Effects for Global Warming:

Higher temperatures: Every continent has warmed substantially since the 1950s. There are more hot days and fewer cold days, on average, and the hot days are hotter.

Heavier storms: The world's atmosphere can hold more moisture as it warms. As a result, the overall number of heavier storms has likely increased since midcentury,

Heat waves: Heat waves have likely become longer and more frequent around the world

Shrinking sea ice: The extent of sea ice in the Arctic has shrunk since 1979, by between 3.5 percent and 4.1 percent per decade, on average. Summer sea ice has dwindled even more rapidly:

Sea-level rise: The pace of sea-level rise has continued to increase in recent decades. Sea-level rise is caused by both the thermal expansion of the oceans — as water warms up, it expands — and the melting of glaciers and ice sheets.

Food supply: A hotter climate can be both good for crops and bad for crops (excess heat can damage plants).

Greenhouse Effect:

The greenhouse effect is the process by which radiation from a planet's atmosphere warms the planet's surface to a temperature above what it would be without its atmosphere. If a planet's atmosphere contains radio active gases (i.e., greenhouse gases) they will radiate energy in all directions. Part of this radiation is directed towards the surface, warming it. The intensity of the downward radiation – that is, the strength of the greenhouse effect – will depend on the atmosphere's temperature and on the amount of greenhouse gases that the atmosphere contains.

Earth's natural greenhouse effect is critical to supporting life. Human activities, mainly the burning of fossil fuels and clearing of forests, have strengthened the greenhouse effect and caused global warming

Ozone Layer

A layer in the earth's stratosphere at an altitude of about 10 km (6.2 miles) containing a high concentration of ozone, which absorbs most of the ultraviolet radiation reaching the earth from the sun.

Causes for Ozone Layer Depletion:

The decrease in ozone concentration in the middle layers of the atmosphere – mainly in the stratosphere – is extremely damaging to life on earth, and is largely caused by emissions of halogenated hydrocarbons produced by man, CFCs, HCFCs, halons, carbon tetrachloride and methyl bromide. For this reason, such substances are commonly referred to as Substances that Deplete the Ozone Layer (ODS).

Effects of Ozone Layer Depletion:

Skin Cancer: Today, it is estimated that skin cancer rates increased due to the decrease in stratospheric ozone (ozone layer Effects on aquatic ecosystems)

The loss of phytoplankton, the basis of the marine food chain, has been observed as the cause of the increase in ultraviolet radiation.

Effects on Animals: For some species, an increase in UV radiation implies the formation of skin cancer. This has been studied in goats, cows, cats, dogs, sheep and laboratory animals and is probably pointing out that this is a common feature of several species.

Plants: In many plants UV radiation can have the following adverse effects: alter its shape and damage plant growth, Reduce tree growth, Change flowering times, Make plants more vulnerable to disease and produce toxic substances.